Arduino-Based Smart Flood and Water-Level Monitoring System with SMS Alert

I. Introduction

The Philippines is highly exposed to tropical cyclones because of its location in the Western North Pacific. According to the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA), an average of 20 tropical cyclones form in or cross the Philippine Area of Responsibility (PAR) every year, with about 8 to 9 crossing the Philippines (PAGASA, n.d.). The peak of the typhoon season occurs from July to October, when nearly 70% of typhoons develop (PAGASA, n.d.).

Tropical cyclones can bring heavy rainfall, strong winds, flooding, storm surges, and landslides, which can affect communities, properties, and daily activities (PAGASA, n.d.). One major concern is flooding because water levels can rise rapidly during heavy and continuous rainfall. This can give residents limited time to prepare or evacuate.

Early warning systems can help address this problem by providing timely information about changing conditions. In the Philippines, the STREAM-EWS initiative has used flood sensors and automated SMS alerts based on water-level thresholds to notify local government units and communities in real time (UNDRR, 2025).

With this in mind, the proposed solution is an Arduino-Based Smart Flood and Water-Level Monitoring System with SMS Alert. The system will monitor water levels and automatically send warnings to registered mobile phones when a predetermined level is reached. Arduino was chosen because it can connect sensors and communication modules and can be programmed to process sensor readings and automatically trigger alerts. Arduino also provides GSM-capable hardware that supports cellular communication (Arduino, n.d.).

II. Problem Description

Flooding caused by tropical cyclones is a recurring concern in the Philippines. With an average of 20 tropical cyclones forming in or crossing the PAR every year and about 8 to 9 crossing the country, communities remain exposed to cyclone-related hazards (PAGASA, n.d.).

One problem during flooding is the difficulty of continuously monitoring the actual water level in a specific location. Residents may rely on manual observation or wait for announcements from authorities. However, when water levels rise quickly, delayed information may also delay preparation and evacuation.

Another problem is that warnings may not immediately reach residents who are away from the monitoring location. This is why a system that can automatically detect rising water levels and send alerts directly to mobile phones can provide an additional source of timely information.

The effectiveness of this approach can be seen in the STREAM-EWS initiative in Mindanao, where flood sensors automatically trigger SMS alerts when predetermined water-level thresholds are reached (UNDRR, 2025). Therefore, the identified problem is the lack of an affordable and automated system that can monitor water levels and immediately notify residents through their mobile phones when flooding reaches a critical level.

III. Proposed Solution

The proposed solution is an Arduino-Based Smart Flood and Water-Level Monitoring System with SMS Alert. The system will use an Arduino microcontroller, an ultrasonic water-level sensor, a GSM module, LED indicators, and a buzzer.

The ultrasonic sensor will measure the water level and send the readings to the Arduino. The Arduino will compare the readings with predetermined thresholds and classify the water level as Normal, Warning, or Critical.

When the water reaches the critical level, the system will activate the buzzer and red LED while the GSM module sends an SMS alert to registered mobile phones. The Arduino MKR GSM 1400, for example, is designed for GSM/3G connectivity and can be used for projects requiring cellular communication (Arduino, n.d.).

IV. Expected Impact

The proposed system can provide real-time and localized flood warnings. By automatically monitoring water levels and sending SMS alerts, residents can receive information even when they are away from the monitoring device. This can give them additional time to prepare, secure their belongings, or follow evacuation instructions.

The system will serve as an additional monitoring and warning tool and will not replace official warnings, forecasts, or evacuation instructions from PAGASA and local authorities.

V. Conclusion

Tropical cyclones continue to pose a significant risk to communities in the Philippines, with an average of 20 tropical cyclones forming in or crossing the PAR every year and around 8 to 9 crossing the country. Flooding caused by heavy rainfall is one of the hazards that can significantly affect vulnerable communities. The proposed Arduino-Based Smart Flood and Water-Level Monitoring System with SMS Alert provides a technology-enabled solution by combining water-level monitoring with mobile communication. Through an Arduino microcontroller, water-level sensor, and GSM module, the system can detect rising water levels and automatically send warnings to registered mobile phones.

References

Arduino. (n.d.). *MKR GSM 1400*. Arduino. https://docs.arduino.cc/hardware/mkr-gsm-1400

Philippine Atmospheric, Geophysical and Astronomical Services Administration. (n.d.). *About tropical cyclone*. DOST-PAGASA. https://www.pagasa.dost.gov.ph/information/about-tropical-cyclone (PAGASA)

Philippine Atmospheric, Geophysical and Astronomical Services Administration. (n.d.). *Tropical cyclone information*. DOST-PAGASA. https://bagong.pagasa.dost.gov.ph/climate/tropical-cyclone-information (PAGASA)

United Nations Office for Disaster Risk Reduction. (2025). *Technology driving innovations for locally tailored tools, real-time data transmission, and interoperable systems for climate resilience and disaster response*. UNDRR. https://www.undrr.org/resource/case-study/technology-driving-innovations-locally-tailored-tools-real-time-data (UNDRR)